Year 13 Maths - Forces in context

Similar questions - Friction on horizontal and inclined surfaces

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A block D of weight 30 N lies on a rough horizontal surface; the coefficient of friction is 0.4. A force of 20 N is applied to D at an angle θ below the horizontal, pushing it forwards and downwards. D remains at rest.

(a) Draw a diagram showing all the forces acting on D. **[1]**

(b) Show that $20\cos\theta - 8\sin\theta \leqslant 12$. **[4]**

(c) Find the smallest possible value of θ, correct to 3 significant figures. **[4]**

2 A block B of weight 20 N lies at rest on a rough plane inclined at θ to the horizontal. A horizontal force of magnitude 5 N, in the vertical plane containing a line of greatest slope, acts on B pushing it towards the plane. The coefficient of friction is 0.5.

(a) Complete a diagram showing all the forces acting on B. **[1]**

(b) Given that B remains at rest, determine the greatest possible value of $\tan\theta$. **[5]**

Question 1

(a) Weight 30 N down, normal reaction R up, friction F opposing the horizontal push, and the 20 N force at θ below the horizontal

(b) Vertically $R = 30 + 20\sin\theta$. Horizontally, friction needed $F = 20\cos\theta$. No sliding: $F \leqslant 0.4R = 12 + 8\sin\theta$, so $20\cos\theta - 8\sin\theta \leqslant 12$

(c) $20\cos\theta - 8\sin\theta = \sqrt{464}\cos(\theta + 21.8^\circ)$. Equality: $\cos(\theta + 21.8^\circ) = \dfrac{12}{\sqrt{464}} = 0.557$, so $\theta + 21.8^\circ = 56.2^\circ$ and $\theta = 34.3^\circ$. The inequality holds for $\theta \geqslant 34.3^\circ$, so the smallest value is 34.3°

Question 2

(a) Weight 20 N down, normal reaction perpendicular to the plane, friction up the plane, and the 5 N horizontal force

(b) Perpendicular to the plane: $R = 20\cos\theta + 5\sin\theta$. Along the plane: $20\sin\theta = 5\cos\theta + F$ and $F \leqslant 0.5R$. So $20\sin\theta - 5\cos\theta \leqslant 10\cos\theta + 2.5\sin\theta$, giving $17.5\sin\theta \leqslant 15\cos\theta$ and $\tan\theta \leqslant \dfrac{6}{7}$. Greatest value of $\tan\theta = \dfrac{6}{7}$